

The boxes a finished unified theory must check

Box a GUT must check

ED status

Minimal foundation (few primitives, no free structure)

✓ strong — reduces to *commitment + participation*; one process primitive does every job

Quantum mechanics from the substrate

✓ reconstructed (Gate 1; residual = Solèr rigor)

Gravity (GR + dark matter + dark energy)

✓ closed (khronometric GR-I..IV + khronon MOND/DE)

QM + gravity unified

✓ distinctive — the *same* arrow is the khronon (gravity) and the pointer basis (QM)

Gauge forces $SU(3) \times SU(2) \times U(1)$

✓ structure derived; ⚠ *why exactly* {1,2,3} is an inherited wall

Charge quantization

✓ from topology (B4 integer winding)

Chirality / parity violation

✓ settled this session — stage built by ED, *which force is chiral* inherited (rep-theory); parity violation now *provably* spontaneous

Matter content (generations, masses, mixing)

⚠ value-inherited; relativistic spinor bridge gated on T4

The constants (α , mass ratios, Λ value)

⚠ inherited by ED's own result (no intrinsic scalar) — not a hole, a structural fact

Internal consistency (anomalies)

✓ conservation (B4) + clean-vector baseline; nontrivial cancellation inherited; one candidate T4-gated

Novel falsifiable predictions

✓ several (Paper_101 inventory) — the *undeniability* axis